

Heart Disease Prediction Using Classification Algorithms

Muhammad Nauman
Department of Informatics
Skövde University
Skövde, Sweden
a24muhna@student.his.se

Abstract—This paper presents a detailed analysis of heart disease prediction using machine learning techniques. It covers data preprocessing, exploratory data analysis (EDA) and the evaluation of multiple classification models. Algorithms such as Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, Support Vector Machine (SVM) and Neural Networks are implemented and assessed using accuracy and Matthews Correlation Coefficient (MCC) metrics. The study highlights the benefits of ensemble learning, with hard voting delivering superior performance. The findings provide actionable insights to improve prediction models and support healthcare decision-making.

Index Terms—Heart Disease Prediction, Machine Learning, K-Nearest Neighbors, Ensemble Learning, Matthews Correlation Coefficient, Data Preprocessing.

I. INTRODUCTION

Heart disease remains one of the leading causes of mortality worldwide, emphasizing the critical need for reliable and early detection methods. Machine learning offers powerful tools for predicting heart disease by utilizing patient data effectively. However, evaluation metrics like accuracy may not always provide a complete picture, especially in datasets with imbalanced classes.

To address this, I have used the Matthews Correlation Coefficient (MCC), which is particularly useful for binary classification with imbalanced data. MCC evaluates all quadrants of the confusion matrix, offering a more balanced view of model performance [5]. By combining accuracy and MCC, I aim to ensure a comprehensive assessment of model effectiveness.

This paper explores various machine learning classification algorithms, emphasizing their performance in predicting heart disease and highlighting the potential of ensemble learning for improved results.

II. DATA PREPROCESSING

The dataset, sourced from the UCI Machine Learning Repository [1], contains 303 records with 14 attributes, including ‘age’, ‘sex’, ‘cholesterol’, ‘thalach’ and the target variable ‘target’. The preprocessing steps include:

- **Handling Categorical Variables:** One-hot encoding was applied to categorical features.
- **Scaling Numerical Features:** Continuous variables were normalized using StandardScaler to ensure uniformity [4].

- **Clean Data:** No missing values were identified in the dataset.

A sample of the preprocessed data is shown in Table I.

TABLE I
SAMPLE PREPROCESSED DATA (SCALED AND ENCODED)

Age	Chol	Thalach	Oldpeak	Sex_1	Cp_2	Target
0.95	-0.26	0.01	1.09	1	0	1
-1.91	0.07	1.63	2.12	1	1	1
-1.47	-0.81	0.98	0.31	0	0	1

III. MODEL EVALUATION

In this study I have evaluated six machine learning algorithms, including Logistic Regression, KNN, Decision Tree, Random Forest, SVM and Neural Network.

A. Randomized GridSearchCV for KNN

The KNN model was optimized using Randomized-SearchCV, which identified the following optimal hyperparameters:

- **n_neighbors:** 24
- **p:** 2 (Euclidean distance)
- **weights:** Uniform

With these parameters, the tuned KNN model achieved an accuracy of 86.89% and an MCC of 0.7395, outperforming the baseline configuration.

B. Ensemble Learning

Ensemble learning combines multiple algorithms to improve overall performance. I have used hard and soft voting ensembles of Logistic Regression, Random Forest and KNN:

- **Soft Voting:** Accuracy of 86.89% and MCC of 0.7371.
- **Hard Voting:** Accuracy of 88.52% and MCC of 0.7707.

Hard voting proved more effective, leveraging the strengths of individual models for robust predictions.

C. Performance Comparison

Table II summarizes the performance metrics, while Figure 2 visualizes the comparison of accuracy and MCC across models.

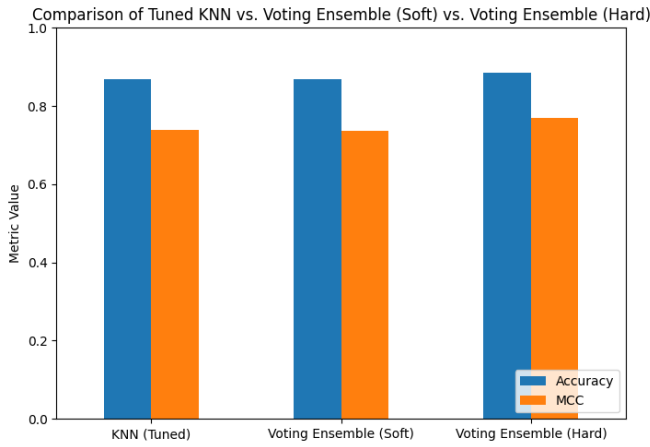


Fig. 1. Comparison of Accuracy and MCC Across KNN(Tuned) and Ensemble model with Voting (Hard and Soft)

TABLE II
MODEL PERFORMANCE METRICS

Model	Accuracy	MCC
Logistic Regression	85.25%	0.70
KNN (Tuned)	86.89%	0.7395
Random Forest	85.25%	0.71
SVM	83.61%	0.67
Neural Network	60.66%	0.31
Ensemble (Hard Voting)	88.52%	0.7707

IV. CONCLUSION

My analysis highlights the potential of machine learning for predicting heart disease. The ensemble learning approach using hard voting outperformed other models, achieving the highest accuracy and MCC. The tuned KNN model also demonstrated strong performance, emphasizing the importance of hyperparameter tuning in classification algorithms.

Future work will explore methods to address class imbalance and integrate additional ensemble techniques for further

improvement.

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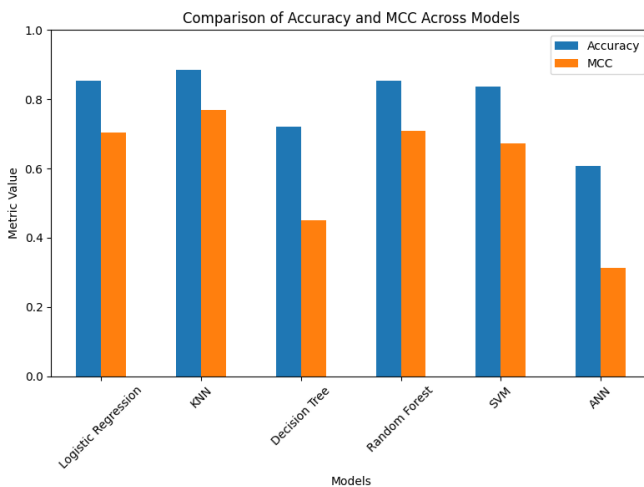


Fig. 2. Comparison of Accuracy and MCC Across Models